S(16): A DETERMINATION WITH EXACT COMPUTER ASSISTANCE

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Theorem. For arbitrary elliptic curves over number fields of exact degree 16,

$$S(16)=\{2,3,5,7,11,13,17,19,23,29,31,37\}.$$

In particular, the largest prime is 37.

The proof uses the corrected dyadic bound, the published results cited below, and exact local certificates. The two additional ingredients are injectivity of rational torsion reduction on $J_1(43)$ at 2, and odd-degree Hecke isogenies on diamond parts of Jacobians at 89 and 97. Failed exploratory tests in this directory are not used as exclusions.

1. OCCURRENCE AND EXACT DEGREE

The known classification S(8)=Primes(23) supplies every prime through 23: extend an octic example by a quadratic extension. Its field then has exact degree 16. For 29,31,37, occurrences.json contains an irreducible polynomial f of degree 16, and x=z,y=g(z)/h(z), where f(z)=0. Set

r=(x^2*y-x*y+y-1)/(x^2*y-x), s=(x*y-y+1)/(x*y),
b=r*s*(r-1), c=s*(r-1), E=[1-c,-b,-b,0,0], P=(0,0).

verify_occurrences.m checks irreducibility, nonsingularity and exact prime order for P. The example at 37 is from van Hoeij's place table; the other two are explicitly verified specializations.

Conversely, a prime in S(16) gives a noncuspidal point on $X_1(p)$ of degree dividing 16. For p>37, the possibilities 1,2,4,8 are excluded by the known classifications. Its degree is therefore exactly 16. Write D for its irreducible rational trace divisor. The S_new(16) criteria apply.

2. THE DYADIC GLOBAL BOUND

We use the earlier corrected dyadic theorem, including the cotangent lattice defect. For $J=J_0(p)$, integral Hecke algebra T, winding element e, I=Ann_T(e), and optimal winding quotient A=J/IJ, the group A(Q) is finite. With $M=H^0(X_0(p)_{\mathbb{Z}_2},\Omega^1)[I]$ and $L=\pi^*\mathrm{Cot}(A_{\mathbb{Z}_2})$, the integral statement is $2M \subset L \subset M$. Equality is not assumed.

Take m=32=2*16 and M0=157. Put

T'_n=sum_(a|n) mu(n/a)*T_a, L_n=T'_(2n)-2*T'_n.

Their first 32 spans agree with that of (T_2-3)T_1,...,(T_2-3)T_32. For p>2*m*M0=10048, and p*u=1 modulo 157, the DKSS intersection formula against {0,a/157} is

epsilon(na)-epsilon(nu/a), 1<=n<=32, 1<=a<=156,

where epsilon indicates the upper half of the nonzero residues mod 157. For every u=1,...,156 the resulting matrix has rank 32 over $\mathbb{F}_2$. prepare_bound.py retains these matrices; verify_auxiliary.py rebuilds and checks every entry and rank independently in Sage.

As in the earlier dyadic theorem, this proves that the cotangent image of the degree-32 Abel map to A at 32*infinity contains $2\mathbb{Z}_2^{32}$. A formal map with pullbacks 2*x_i plus higher terms is injective on the formal neighbourhood modulo 4: for minimum coordinate valuation k>=2, a linear term has valuation k+1<2k. If a degree-16 divisor reduces to 16*c modulo 2, its elementary symmetric polynomial is 1+2B(t). Its double has polynomial 1+4(B+B^2), so agrees with 32*c modulo 4. The torsion in the kernel of reduction of an abelian scheme over $\mathbb{Z}_2$ is killed by 2. Thus the doubled-divisor argument applies despite the cotangent defect.

If p>(2^8+1)^2=66049, Hasse’s bound excludes good reduction at every place above 2. Additive reduction cannot support this prime torsion. The nonsplit multiplicative and identity-component bounds also fail; the point must lie in a split multiplicative component group. After Atkin--Lehner on X_0(p), the trace reduces to 16*infinity, and the preceding argument excludes it. Since 10048<66049, we have p<=66049. The full local lattice argument and references are in the previously established dyadic theorem, reproduced in the earlier proof packets.

3. THE GLOBAL REDUCTIONS, INCLUDING 127 AND 193

Derickx--Stoll, arXiv:2505.14109v1, Table 2, Proposition 4.1, Remark 4.4 and Corollary 7.5 are used only for p<100000. The two CM alternatives p=2*16+1 or p=3*16+1 are not primes. The exact arithmetic ledger applies Corollary 7.5 and, where applicable, Proposition 4.1 with ell=2,a=2,n=5. It leaves the 20 primes
41,43,47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113,127,193.

For 127, we use Booker--Lee--Strombergsson, Theorem 1.1: the Selberg bound lambda_1>=1/4 is proved for Gamma_1(N), N<=880. The Li--Yau/ Abramovich inequality then gives
gon(X_1(127)) >= (127^2-1)/192=84.
The prime 127 is non-strange. Proposition 4.1 applies with ell=2,a=2, n=5 and 5*16=80<84; Remark 4.4 gives its rational2-torsion hypothesis. The diamond 2 has order 7 modulo sign and acts freely on characteristic-zero noncuspidal points. Invariance would require 7|16. This excludes 127. This uses the proved small-level spectral result, not an assumed Selberg conjecture.

For 193, Corollary 7.5 requires a noncuspidal quadratic point on
Y=X_1(193)/H, with H the order-8 subgroup of the diamond group.
The degree of Y above X_0(193) is 12. Its rational cusps form one orbit of size 12; the other cusps have closed degree 6 at 2. The complete census on X_1(193) through degree 16 contains no noncuspidal point with j-invariant of degree at most 2 over F_2. Thus a quadratic point on Y cannot have noncuspidal reduction: a point above it on X_1(193) would have degree at most 16 and j-degree at most 2. Its reduction is therefore supported on rational cusps of Y.

On the genus-177 curve Y, quotient193.m and export_new_finite.m verify an integral Hecke polynomial whose operator image lies in the winding module. The computation uses projected period coordinates, with Hecke operator Transpose(DualHeckeOperator), and checks compatibility with the ambient projection. The T_2 winding span has dimension 159. Its finite-image polynomial, recorded in finite193.m, is composed with the good-prime torsion annihilator at 3. On degree-2 cusp prefixes the actual integral endomorphism cotangent matrices have rank 2, for all 12 normalized patterns. Formal immersion makes any such quadratic divisor equal to its rational cuspidal lift. This contradicts noncuspidality and excludes 193. The two new winding inclusions are checked independently in Sage.

4. THE LOCAL CENSUS, FORMAL IMMERSION AND ORDINARY OBSTRUCTIONS

The remaining 18 levels are
41,43,47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113.
All computations at these levels use the full point-level modular curve. The census enumerates all j-invariants and twists over F_(2^d), 1<=d<=16. When full rational p-torsion occurs it enumerates all p+1 cyclic lines,

rather than just one. Tate coordinates determine exact closed degrees,
closed diamond orbits and duplicate keys. All previous ordinary jets
are reused only after unique exact matching to this new census.

For rational-cusp reductions, actual integral endomorphisms with finite
rational image are composed with an odd-prime rational-torsion
annihilator, giving maps which kill the whole rational Jacobian group.
Fricke transports the cotangent test from the mu-level convention to
the geometric point convention. The chosen diamond indices are
 41:20,43:21,47:23,53:13,59:29,61:15,67:11,71:7,73:12,79:39,
 83:41,89:11,97:8,101:5,103:17,107:53,109:54,113:56.
At 43 the finite polynomial is T_2+2, and maps with q=3,5,7 are combined.
The other selected finite-image polynomials are the previously certified
ones. The rank-zero levels 41,47,59,71 allow any integral polynomial;
at 107,109,113 a diamond difference kills all positive-rank factors.
At 61,97,101 the selected indices omit the respective strange character
orders 6,12,10. The non-strange cases have positive-rank factors only
in their trivial-character part. The winding certificates use the correct
dual period-coordinate action throughout.

Every normalized rational cusp multiplicity pattern of degree 16 passes.
The core uses 16 rows at the first cusp and six at every other cusp.
The remaining patterns are precisely
 8+8, 9+7, 8+7+1, 7+7+2, 7+7+1+1.
All relative cusp positions are checked. Complete prefix enumeration,
or a complete binary search in the relation space, checks the core;
independent dynamic-programming counts check the enumeration totals.
Sage separately checks the full ranks, core ranks and relation bases.
These are integral endomorphism tests, so no optimal-quotient cotangent
saturation is being assumed at 2.

For ordinary noncuspidal support, write sigma for the recorded primitive
geometric diamond, and A3=T_3-1-3*delta_3. The operator A3 kills rational
torsion and each rational cusp divisor. We apply alpha*A3, with
 41,47,59,71: alpha=1;
 61: alpha=sigma^6-1;
 97: alpha=sigma^12-1, with (sigma-1)*Phi12(sigma) on the three degree10/14 closed orbits
of size24;
 101: alpha=(sigma-1)*(sigma^5+1);
 all other levels: alpha=sigma-1.
These alpha have finite rational image, by the rank-zero and strange-
character results. Diamond inversions in the input convention do not
change the character orders to be killed.

The tests construct actual Hecke targets and canonical jets from elliptic
curves and torsion points. Products of sections are checked by q-expansions
beyond the applicable Sturm bound. The section space is H^0(K^2) at
41,43,53 and H^0(K) elsewhere. If U-V were principal, choose nonzero
F,H in R(-U). With div(f)=U-V, S=F/f is a nonzero section in R(-V), and
H*S is divisible by F in the product section space. The recorded linear
conditions on S would have a nonzero solution. Full matrix rank therefore
proves nonprincipality. The argument also holds if F vanishes at some
negative support points.

There are 4521 ordinary source orbits, of which 4515 pass directly. The
six exceptions are
 41: degree 14, orbit 10;
 43: degree 12, orbit 7; degree 14, orbit 21; two degree-15 orbits of size 7;
 113: degree 8, orbit 7.

They are resolved below.

Two ordinary points can have types 7+7,7+8,7+9,8+8. The 137 degree-14 pair certificates and 254 degree-15 pair certificates remain valid with a rational cusp complement, because the operator kills that complement. The 561 new total-degree-16 pair configurations give 558 full-rank matrices. Their only three exceptions are two translates of the degree-8 orbit of size 7 at 113; these are resolved in Section 8.

Mixed tests give another 34 full-rank matrices. At 41, use (sigma^2-1)A3 on ordinary degrees 8,9,10,11; it kills a supersingular degree-5 contribution. At 43, use (sigma^3-1)A3 on degrees 8,9; it kills either a supersingular degree-7 point or a nonrational degree-7 cusp. At 73, use (sigma^4-1)A3 on degrees 7,10 for the degree-6 supersingular contribution, and on degree 7 for the degree-9 nonrational cusp. The remaining mixed case, at 113, is handled by an involution below. Sage independently checks all 5498 principal-divisor obstruction matrices used here, including the borrowed pairs.

5. TWO STABILIZER LEMMAS

Let C be the rational cusp lift of the cusp part of the reduction of D, of degree c, and let h fix its nonrational reduction. If (h-1)J(Q) is finite, then y=[h(D-C)-(D-C)] is rational torsion reducing to zero. To justify torsion for divisors of nonzero degree, subtract a rational cusp as base point; the resulting extra cusp difference is torsion by Manin--Drinfeld. The formal-group lemma at 2 gives 2y=0.

If rational torsion reduction is injective, then y=0. More generally the same holds when h has odd order and acts trivially on J(Q)[2], by the norm relation. If gon_Q(X)>16+c, principality then forces h(D-C)=D-C as divisors. Noncuspidal and cuspidal supports are disjoint, so hD=D and hC=C. Since diamonds act freely on rational cusps, ord(h)|c. Without y=0, the doubled inequality gon_Q(X)>2*(16+c) gives the same conclusion from 2y=0.

An additional way to prove y=0 is used at 89 and 97. Let h have odd order r and B=(h-1)J, and suppose a good-prime torsion annihilator A3 has odd degree as an isogeny on B. Since N_h*y=0 and 2y=0, we have y in B: r-N_h is divisible by h-1 in Z[h], and r*y=y. Thus y is a rational2-torsion point on B killed by the odd-degree isogeny A3, so y=0. No choice of an integral homology lattice is needed for this parity test: the degree is the determinant on rational Betti homology. Both signs are computed and independently checked, as described below. The computed matrix T_3-1-3*<3> is congruent modulo 2 to T_3-<3>-3. Fricke transport multiplies the latter annihilator by a diamond. Thus the odd-degree conclusion applies to the geometric point-level torsion annihilator as well, regardless of the point/mu convention.

6. INJECTIVITY OF RATIONAL TORSION REDUCTION AT LEVEL 43

We use Varivoda, Gonality and quintic points on intermediate modular curves, Theorem D, which gives gon_Q(X_1(43))=24. The code and model are pinned at commit 66c5f61e9ead7ea430f6d2c8538237e01a83e652 in the supplied checkout intermediate-modular-curves. The genus-9 quotient Y=X_1(43)/<delta_2>, with delta_2 of order 7 modulo sign, is the curve 43.132.9.a.2 identified in that paper, Section 4.3.

Here is the additional torsion argument, with fresh arithmetic checks.

Up to Q-isogeny, J_Y has factors of dimensions 1,2,2,4 with exact
L-ratios 0,2/7,3/16,1/361. Its positive-rank elliptic factor is isogenous
to E: y^2+y=x^3+x^2 (43a1). This rational isogeny class has one curve,
and its rational torsion is trivial. Thus the actual elliptic subvariety
A of J_Y also has A(Q)[2]=0. The actual quotient B=J_Y/A has rank zero.
Its good-prime point counts at 3,5,23 are, respectively,
 3192,80864,86398048604,
whose gcd is 532, of 2-adic valuation 2. Point counts are invariant under
Q-isogeny. Hence J_Y(Q)[2] injects into B(Q), and has at most four elements.
torsion43_upper.m checks all these factor, L-ratio, count and elliptic
isogeny assertions directly.

Let C1,C2,C3 be the three rational cusps, cyclically permuted by the
remaining diamond group. The rational divisor 38(C2-C1) is principal,
as in the cited paper’s explicit torsion calculation; torsion43_Q_first.m
rechecks this one needed relation. Applying the cyclic deck action shows
that 38(C3-C1) is principal too. Therefore
 x1=19[C2-C1], x2=19[C3-C1]
are rational2-torsion. On the smooth reduction at 2, fresh exact
Riemann--Roch calculations give l(x1)=l(x2)=l(x1+x2)=0 and l(2x1)=
l(2x2)=l(2x1+2x2)=1, with each expression understood as its displayed
degree-zero divisor. Thus all three nonzero classes remain nonzero.
They generate all of J_Y(Q)[2], and its reduction is injective.

The canonical integral model used for this check has absolutely
irreducible special fibre of degree 16 and arithmetic and geometric
genus 9, with the generic canonical Hilbert polynomial, hence
is smooth. The three cusp coordinates are reduced from the actual
rational cusps on the canonical model. Selecting arbitrary branches on
the singular plane projection would not give this check.

On the actual integral sign0 homology of JOne(43), the q=3,5,7
annihilator matrices bound J_1(43)(Q)[2] by a space on which delta_2
acts trivially; two43_h7_data.m records the matrices. If x is rational
torsion reducing to zero on J_1(43), then 2x=0. For the degree-7 quotient
f:X_1(43)->Y, f_*x=0 by injectivity on J_Y(Q)[2], and
 f^*f_*x=sum_(h in <delta_2>) h*x=7x=x.
Thus x=0. Rational torsion reduction at 2 is injective on J_1(43).

For each ordinary exception of degree 12 or 15, the order-3 diamond
37=3^7 fixes the nonrational reduction. The cusp complement degrees are
4 and 1. Since 24>16+4 and 24>16+1, Section 5 forces hC=C, contradicting
3 not dividing 4 or 1. This resolves all three such ordinary exceptions.

At 43, the single degree-7 supersingular reduction and the single
nonrational degree-7 cusp were already excluded by the audited native
alpha*A3 certificates, valid with any rational cusp complement.
Every nonordinary support of total degree 14 (one supersingular point
of degree 14, two degree-7 supersingular points, two degree-7 nonrational
cusps, or one of each) is fixed by the order-7 diamond 2. Its complement
has degree 2, and 24>18. Section 5 excludes it. The mixed ordinary cases
were covered in Section 4.

7. THE PRIMES 53, 89 AND 97

At 53, the unique supersingular point has degree 13, so c=3. The
order-13 diamond 4 fixes it. The actual JOne(53) calculation proves
J_1(53)(Q)[2]=0. We strengthen the previous gonality bound to
 gon_Q(X_1(53))>=20.

There are 26 rational cusps over F_2 and no other places through degree 8. A function of degree at most 19 has a rational fibre containing at least 9 distinct rational points; move that fibre to infinity. Its pole divisor is bounded by C_all+E, with C_all the sum of all 26 rational cusps and E effective of degree at most 10. It suffices to pad E to degree 10. Exact canonical evaluation has rank 26 at C_all and adds 10 independent conditions for every rational-cusp E. The normalized prefix search checks all 12,982,006 patterns. The other possibilities are one degree-9 ordinary point plus one rational cusp, or one degree-10 ordinary point. All 78 and 130 diamond translates, respectively, are checked, with every possible rational cusp in the former case. In every case l(C_all+E)=1. This proves the bound. Since 20>16+3, Section 5 would require 13|3, excluding 53.

At 89, the only nonordinary case is a degree-11 cusp, with c=5, fixed by the order-11 diamond 2. In its diamond part B=(delta_2-1)J_1(89), of dimension 260, A3 has odd determinant on both Betti signs. The two identical determinants are recorded in odd_anti_data.m and independently checked in Sage. Section 5 therefore gives y=0. The ordinary gonality bound 325*(89^2-1)/65536>21=16+5 then forces 11|5, a contradiction.

At 97, the only nonordinary case is one supersingular closed point of degree 16, whose closed diamond orbit has size 2. Let Delta be the order-48 diamond group generated by sigma=delta_5, and H4=<sigma^12>, H8=<sigma^6>. The spectral bound from Section 3 gives

 gon(X_1(97)) >= (97^2-1)/192=49.

The group (h-1)J_1(97)(Q) is finite for h in H4: the only nontrivial strange character order at 97 is 12. The supersingular closed point is H4-invariant. With c=0, the doubled stabilizer bound 32<49 therefore forces D to be H4-invariant. The possible order-2 elliptic stabilizer would require j=1728; cm_small.m excludes that j in degree 16. Thus D descends to an irreducible quartic divisor on X_1(97)/H4.

Put Z=X_1(97)/H8. It has genus 41, and its rational gonality is at least 7, since a degree at most 6 map would compose to a degree at most 48 map on X_1(97). The image D_Z is irreducible of degree m=2 or 4: degree 1 would give a rational noncuspidal point on X_0(97). Let h be the order-3 diamond on Z induced by sigma^2. It fixes the reduction of D_Z, because sigma^2 fixes the original supersingular closed point. More explicitly, f_*D=(16/m)D_Z, and specialization of this identity proves invariance of the reduction in the torsion-free divisor group.

The group (h-1)J_Z(Q) is finite: the index 6 omits the strange character order 12. The class y=[hD_Z-D_Z] is rational torsion reducing to zero, so 2y=0. In B=(h-1)J_Z, of dimension 28, the q=3 annihilator has

 determinant 30770604725769

on each Betti sign. Its isogeny degree is therefore odd. Section 5 gives y=0, and gon_Q(Z)>=7>m forces hD_Z=D_Z. Since 3 does not divide 2 or 4, h fixes every geometric point of D_Z. Such a noncuspidal fixed point must have j=0: an elliptic stabilizer whose image in Delta/H8 has order 3 must come from an order-3 automorphism modulo sign. Hence the original point D has j=0. The explicit Frobenius certificate cm_small.m rules out j=0 and 97-torsion over degree 16. This excludes 97.

For clarity, the Frobenius certificates for rational j use a fixed E/Q with that j and w=#Aut(E_Qbar). A point above j is an Aut(E)-orbit of nonzero p-torsion. If Frobenius has an orbit of length k on these moduli points, then Frob^(w*k) fixes a nonzero p-point (Galois acts on the automorphisms by inversion or trivially). For a good prime q the characteristic polynomial is x^2-a_q*x+q. The script checks that the

companion matrix to this polynomial has F^(w*k)-1 invertible modulo p
for every 1<=k<=16. Thus there is no closed point of degree at most 16
above that j. At 97 the chosen (j,q,a_q,w) are (0,5,0,6) and
(1728,3,0,4). The checks concern moduli points and cover twists as well.

8. THE OTHER EXCEPTIONAL SUPPORTS AND THE KNOWN LIFTS

At 41, the earlier independently checked native certificates exclude
one or two degree-5 supersingular points, a degree-10 nonrational cusp,
and their rational cusp complements. The degree-15 computation adds
both triple types 3P and 2P+Q, and the degree-10 cusp plus either degree-5
supersingular point. Their operators kill rational cusp contributions,
so the same tests apply in total degree 16. The mixed ordinary cases
are covered by the 23 new matrices of Section 4.

For the remaining ordinary degree-14 orbit at 41, a known irreducible
characteristic-zero degree-14 lift is checked directly in Tate coordinates.
The combined cotangent maps with q=3,5,7 have rank 16 for all 210
rational cusp complements of degree 2, including repeated cusps.
Formal immersion forces D to equal this reducible lift plus its cusp
complement. This contradicts irreducibility.

The ordinary degree-14 orbit at 43 is similarly the reduction of the
non-CM degree-14 place in van Hoeij’s table. lift43_14.m verifies the
number-field example, exact order 43 and its direct Tate-coordinate match.
All 231 rational cusp complements of degree 2 give rank 16 for the
combined finite-image annihilator maps. Formal immersion again forces
a reducible lift. Sage independently checks all 210+231 matrices.

At 61, the remaining supersingular degree-15 point has complement c=1.
Diamond 3 has order 5 and finite rational difference image. The actual
JOne(61) certificate gives J_1(61)(Q)[2]=0. Since the gonality bound is
>17, Section 5 would require 5|1, a contradiction.

At 73, use the order-9 diamond 2, which acts trivially on the certified
upper bound for J_1(73)(Q)[2]. It fixes every relevant degree-6
supersingular point and the degree-9 nonrational cusp. The complements
for one supersingular point, two such points, one nonrational cusp,
and a supersingular point plus that cusp have degrees 10,4,7,1.
The gonality bound is >26, and none of those degrees is divisible by 9.
The norm form of Section 5 excludes them. The mixed ordinary cases,
including a degree-9 cusp plus an ordinary degree-7 point, are in Section 4.

At 109 the degree-9 and degree-12 supersingular points are fixed by the
order-18 diamond 23. Their complements have degrees 7 and 4. The doubled
bounds 2*(16+7)=46 and 2*(16+4)=40 are below the gonality bound, while
18 divides neither complement degree.

At 113 the degree-7 supersingular points, pairs of them, and degree-14
nonrational cusps are fixed by the order-14 diamond 2. Complements have
degrees 9 or 2, neither divisible by 14. The doubled stabilizer inequalities
hold. A degree-7 supersingular point plus an ordinary degree-8 point has
c=1. The involution 98 fixes both supports, for both ordinary orbits of
sizes 7 and 28. Since gonality>34, it would fix a rational cusp, impossible.

It remains to treat the ordinary degree-8 orbit of size 7 with eight
rational cusps, and the three surviving pairs of its translates. The
order-8 diamond 40 fixes the nonrational supports. The doubled bounds
are at most 48, below the gonality bound. Thus D is invariant under this

order-8 group. If an elliptic stabilizer occurs then j=1728, already excluded by cm_small.m. Otherwise D descends to a quadratic point on the quotient by this group, and hence to a point of degree at most 2 on X_0(113). There are no rational noncuspidal points. The complete quadratic-point classification of Adzaga--Keller--Michaud-Jacobs--Najman--Ozman--Vukorepa, Table 14, gives only the rational CM j-invariants
287496,1728,16581375,-3375,-32768,-262537412640768000,8000.
For each, cm_small.m supplies a good-prime Frobenius certificate as explained in Section 7 excluding all degrees at most 16. This resolves both the single ordinary exception and all pair exceptions at 113.

9. COMPLETENESS AND REPLAY

The nonordinary closed-point degrees which fit are
41: SS5, NC10;
43: SS7, SS14, NC7;
53: SS13; 61: SS15; 73: SS6, NC9; 89: NC11; 97: SS16;
109: SS9,SS12; 113: SS7,NC14.
All other closed support points are ordinary or rational cusps. The minimum ordinary degree is at least 7, so there are at most two ordinary support points. The cases in Sections 4 and 8 cover all their possible pairs. With one ordinary point there is at most one nonordinary point; all mixtures are the ones listed in Sections 4 and 8. Without ordinary points, the above degrees allow at most three degree-5 points at 41, at most two degree-6 or degree-7 points at 73,43,113, and only the other singles or the expressly listed SS/NC pairs. Thus every possible degree-16 reduction is covered.

verify_ranks.py independently checks all 5498 obstruction matrix ranks using Sage. verify_auxiliary.py independently rebuilds the 156 dyadic matrices and checks the formal-immersion matrix ranks and relation bases, the known-lift matrices, the four signed odd-isogeny determinants, finite-image inclusions, rational2-torsion bounds and CM Frobenius tests. The genus-53 rational and nonrational searches retain their full counts. verify_coverage.py checks all prime and support-case ledgers. The final coverage has no unresolved primes and matching occurrence and upper sets. No separate agent audit of this S(16) proof is claimed.

Run the Magma inputs from the workspace root and save their outputs to the corresponding .log files. Magma V2.30-1 and Sage Python were used. The scripts use the installed mdmagma library at their AttachSpec paths. The earlier native certificates and the degree-14 and degree-15 pair certificates are intentionally reused. Exploratory files native43.m, native53.m, sieve5.py, sieve11.py, anti97_full.m, anti97_middle.m, two97_data.m and the occurrence search at 43 are not proof inputs.

References

Derickx--Kamienny--Stein--Stoll, Torsion points on elliptic curves over number fields of small degree, arXiv:1707.00364, and the degree-8 classification of Khawaja.
Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109v1, especially Table 2, Proposition 4.1, Remark 4.4 and Corollary 7.5.
Booker--Lee--Strombergsson, Twist-minimal trace formulas and the Selberg eigenvalue conjecture, Theorem 1.1, https://doi.org/10.1112/jlms.12349 .
Varivoda, Gonality and quintic points on intermediate modular curves, Theorem D and Section 4.3, https://hexagonmath.org/2610.00219v1 .
Adzaga--Keller--Michaud-Jacobs--Najman--Ozman--Vukorepa, Computing

quadratic points on modular curves X_0(N), Table 14,
https://doi.org/10.1090/mcom/3902 , arXiv:2303.12566v2.
van Hoeij, Low degree places on the modular curve X_1(N),
arXiv:1202.4355, and the explicit place table supplied with the workspace.
The earlier corrected dyadic theorem, including its lattice argument.

CORRECTION TO THE ORDINARY LEVEL97 CERTIFICATES (29 October 2026)

The original code polynomial sigma^6-sigma^4-sigma^2+1 does not
annihilate the strange order12 character. It has been replaced by
sigma^12-1 and, on the three closed orbits of size24 in degrees10/14,
(sigma-1)*Phi12(sigma). All ordinary level97 residue orbits through degree17 have
been recomputed with the corrected map and independently checked in
Sage. The relevant lower-degree matrices here are exact subsets of
that universal certificate. The other level97 arguments are unchanged.
See outputs/S97_correction.pdf and work/degree17/full97/sieve.json.